

GPT-4o

Q: The guest will sit on the right side of the table, near the yellow cup. Where should we serve the apple?



A: Apple is at $[445, 535]$, , and should be placed at $[450, 150]$.

GPT-4o with Autonomous Imagination

Q: The guest will sit on the right side of the table, near the yellow cup. Where should we serve the apple?



Creating the imagination space.
The apple is right at the cursor.

Reasoning in Imagination Space



I have located the apple at $[450, 450]$, .

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Reasoning in Imagination Space



This location looks better, but I will
try to find a better location.

...

Reasoning in Imagination Space



A: I think this is a nice spot.
Target location: $[705, 376]$.